Technology Stack

Date	03 October 2026
Team ID	TEAM-ID-TO-BE-FILLED
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, vanilla JavaScript, Jinja2 templates	No build step, fast to load, responsive; Jinja2 shares one base layout across 8 pages; AI text is rendered with `textContent` to prevent injection
2	Backend / Server-Side	Python 3, FastAPI, Uvicorn, Pydantic	Async-capable, automatic request validation and OpenAPI docs at `/docs`, team knows Python
3	Database / Data Storage	SQLite (standard-library `sqlite3`)	Zero-setup relational store for users and saved plans; schema in `database.py`; easily replaced by PostgreSQL
4	Cloud / Hosting / Deployment	Local Uvicorn server, Docker (`Dockerfile` provided)	Simple to run for the demo; container image deployable to Render, Railway or any cloud VM
5	Version Control & CI/CD	Git and GitHub, pytest	Collaboration for five members; automated test suite (43 tests) ready to plug into GitHub Actions
6	Third-Party APIs / Other Tools	Google Gemini API (`google-genai` SDK), Pillow, bcrypt, python-jose (JWT), python-dotenv	Gemini generates budget-aware text and reads outfit images (multimodal); Pillow validates uploads; bcrypt and JWT secure accounts; dotenv keeps keys out of code